ADVANCED JAVA PROGRAMMING LAB

(Course Code: 23UPCSC1L06)

A programming laboratory record submitted to Periyar University, Salem

In partial fulfilment of the requirements for the degree of

MASTER OF COMPUTER APPLICATION

By

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Salem – 636 011.

(OCTOBER - 2025)

CERTIFICATE

This is to certify that the Programming Laboratory entitled "**ADVANCED JAVA PROGRAMMING LAB (23UPCSC1L06)"** is a Bonafide record work done by Mr. /Ms._____ Register No: _____ as partial fulfilment of the requirements for the degree of Master of Computer Applications, in the Department of Computer Science, Periyar University, Salem, during the Academic Year 2025-2026.

Staff In-charge Head of the Department

Submitted for the practical examination held on……..……

Internal Examiner External Examiner

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SOURCE CODE:

Student.java:

```
import java.lang.*;
import java.io.*;
interface Exam {
    void percent_cal();
}
class Student {
    String name;
    int roll_no, mark1, mark2;
    Student(String n, int r, int m1, int m2) {
        name = n;
        roll_no = r;
        mark1 = m1;
        mark2 = m2;
    }
    void display() {
        System.out.println("====");
        System.out.println("Student Details");
        System.out.println("====\n");
        System.out.println("Name of Student: " + name);
        System.out.println("Roll No. of Student: " + roll_no);
        System.out.println("Marks of Subject 1: " + mark1);
        System.out.println("Marks of Subject 2: " + mark2);
    }
    public static void main(String[] args) {
        Result R = new Result("Mr.X", 12, 93, 84);
        R.display();
        R.percent_cal();
    }
}
class Result extends Student implements Exam {
```

```
    Result(String n, int r, int m1, int m2) {
        super(n, r, m1, m2);
    }
    public void percent_cal() {
        int total = (mark1 + mark2);
        float percent = (total * 100) / 200;
        System.out.println("Percentage: " + percent + "%");
    }
    void display() {
        super.display();
    }
}
```

OUTPUT:

```
Student Details
====

Name of Student: Mr.X
Roll No. of Student: 12
Marks of Subject 1: 93
Marks of Subject 2: 84
Percentage: 88.0%
```

RESULT:

SOURCE CODE:

Exception.java:

```
class exception {
   public static void main(String args[]) {
      try {
         int a = 30, b = 0;
         int c = a / b;
         System.out.println("Result = " + c);
      } catch (ArithmeticException e) {
         System.out.println("Can't divide a number by 0");
      }

      try {
         int num = Integer.parseInt("akki");
         System.out.println(num);
      } catch (NumberFormatException e) {
         System.out.println("Number format exception");
      }

      try {
         int a[] = new int[5];
         a[6] = 9;
      } catch (ArrayIndexOutOfBoundsException e) {
         System.out.println("Array Index is Out Of Bounds");
      }

      try {
         Object[] objArray = new Integer[10];
         objArray[0] = "String";
      } catch (ArrayStoreException e) {
         System.out.println("ArrayStoreException: " + e.getMessage());
      }

      try {
         String a = null;
```

```
            System.out.println(a.charAt(0));
        } catch (NullPointerException e) {
            System.out.println("NullPointerException: " + e.getMessage());
        }
    }
}
```

OUTPUT:

```
_e42e8491/bin exception
Can't divide a number by 0
Number format exception
Array Index is Out Of Bounds
ArrayStoreException: java.lang.String
NullPointerException: null
```

RESULT:

SOURCE CODE:

AccessModifier.java:

```
class Employee {
    public String name;
    protected int employeeId;
    private double salary;
    public Employee(String name, int employeeId, double salary) {
        this.name = name;
        this.employeeId = employeeId;
        this.salary = salary;
    }
    public void displayDetails() {
        System.out.println("Name: " + name);
        System.out.println("Employee ID: " + employeeId);
        System.out.println("Salary: $" + salary);
    }
    protected void calculateBonus() {
        double bonus = salary * 0.1;
        System.out.println("Bonus: $" + bonus);
    }
}
class Manager extends Employee {
    private String department;
    public Manager(String name, int employeeId, double salary, String department) {
        super(name, employeeId, salary);
        this.department = department;
    }
```

```
    public void displayDetails() {
        super.displayDetails();
        System.out.println("Department: " + department);
    }
}
class Staff extends Employee {
    private int workingHours;
    public Staff(String name, int employeeId, double salary, int workingHours) {
        super(name, employeeId, salary);
        this.workingHours = workingHours;
    }
    public void displayDetails() {
        super.displayDetails();
        System.out.println("Working Hours: " + workingHours);
    }
}
public class AccessModifier {
    public static void main(String[] args) {
        Manager manager = new Manager("Mr.X", 101, 50000.0, "Sales");
        Staff staff = new Staff("Mr.Y", 202, 60000.0, 50);
        System.out.println("Manager Details:");
        manager.displayDetails();
        manager.calculateBonus();
        System.out.println();
        System.out.println("Staff Details:");
        staff.displayDetails();
        staff.calculateBonus();}}
```

OUTPUT:

```
Manager Details:
Name:Mr.X
Employee ID:101
Salary:$50000.0
Department:Sales
Bonus:$5000.0

Staff Details:
Name:Mr.Y
Employee ID:202
Salary:$60000.0
Working Hours:50
Bonus:$6000.0
```

RESULT:

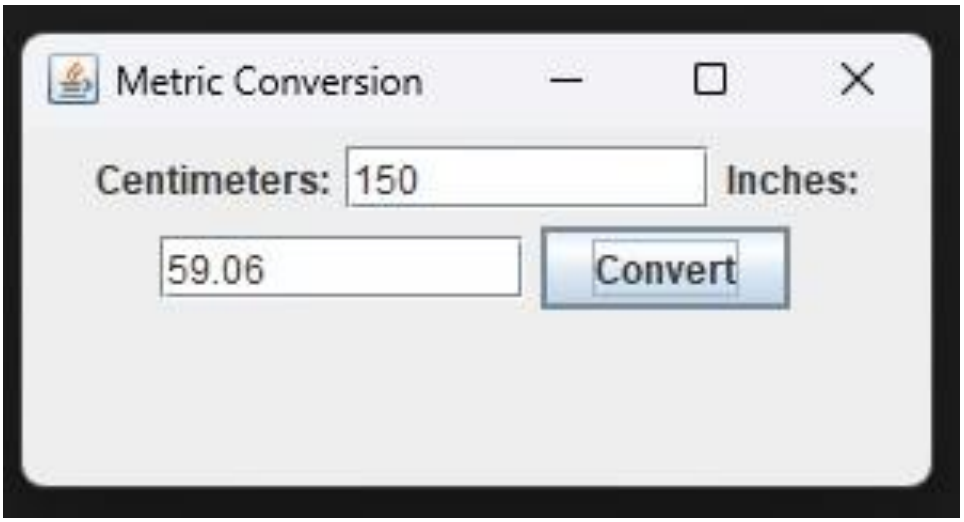
SOURCE CODE:

MetricConversionApp.java:

```
import javax.swing.*;
import java.awt.*;
public class MetricConversionApp extends JFrame {
    private JLabel cmLabel;
    private JTextField cmTextField;
    private JLabel inchLabel;
    private JTextField inchTextField;
    private JButton convertButton;
    public MetricConversionApp() {
        setTitle("Metric Conversion");
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setSize(300, 150);
        setLayout(new FlowLayout());
        cmLabel = new JLabel("Centimeters:");
        cmTextField = new JTextField(10);
        inchLabel = new JLabel("Inches:");
        inchTextField = new JTextField(10);
        convertButton = new JButton("Convert");
        convertButton.addActionListener(e -> {
            try {
                double centimeters = Double.parseDouble(cmTextField.getText());
                double inches = centimeters / 2.54;
                inchTextField.setText(String.format("%.2f", inches));
            } catch (NumberFormatException ex) {
                JOptionPane.showMessageDialog(
                    null,
```

```
                "Invalid input. Please enter a valid number."
            );
        }
    });
    add(cmLabel);
    add(cmTextField);
    add(inchLabel);
    add(inchTextField);
    add(convertButton);
  }
  public static void main(String[] args) {
    SwingUtilities.invokeLater(() -> {
      MetricConversionApp app = new MetricConversionApp();
      app.getRootPane().setDefaultButton(app.convertButton);
      app.setVisible(true);
    });
  }
}
```

OUTPUT:



RESULT:

SOURCE CODE:

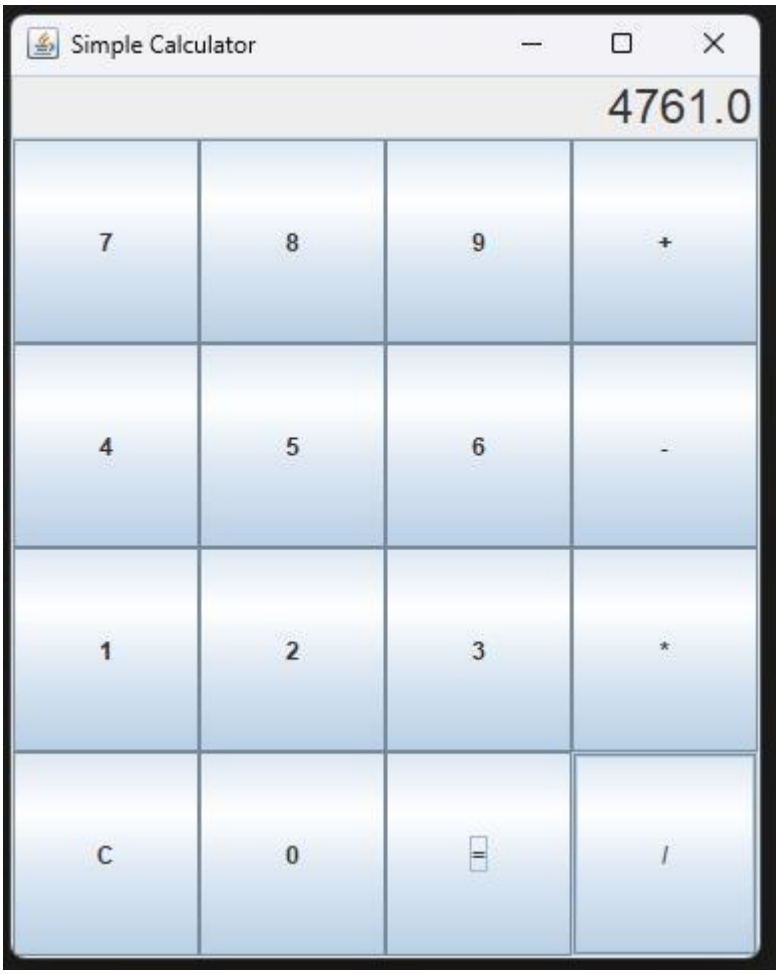
Calc.java:

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
public class calc extends JFrame implements ActionListener {
    private JTextField textField;
    private JButton[] buttons;
    private String input = "";
    private double num1, num2, result;
    private char operator;
    public calc() {
        setTitle("Simple Calculator");
        setSize(300, 400);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLayout(new BorderLayout());
        textField = new JTextField(10);
        textField.setFont(new Font("Arial", Font.PLAIN, 24));
        textField.setHorizontalAlignment(JTextField.RIGHT);
        textField.setEditable(false);
        JPanel panel = new JPanel(new GridLayout(4, 4));
        String[] buttonLabels = {
            "7", "8", "9", "+",
            "4", "5", "6", "-",
            "1", "2", "3", "*",
            "C", "0", "=", "/"
```

```
        };
        buttons = new JButton[buttonLabels.length];
        for (int i = 0; i < buttonLabels.length; i++) {
            buttons[i] = new JButton(buttonLabels[i]);
            buttons[i].addActionListener(this);
            panel.add(buttons[i]);
        }
        add(textField, BorderLayout.NORTH);
        add(panel, BorderLayout.CENTER);
        setVisible(true);
    }
    public void actionPerformed(ActionEvent e) {
        String command = ((JButton) e.getSource()).getText();
        if (command.matches("[0-9]")) {
            input += command;
            textField.setText(input);
        } else if (command.matches("[+\\-*/]")) {
            num1 = Double.parseDouble(input);
            operator = command.charAt(0);
            input = "";
        } else if (command.equals("=")) {
            num2 = Double.parseDouble(input);
            switch (operator) {
                case '+':
                    result = num1 + num2;
                    break;
                case '-':
```

```
                result = num1 - num2;
                break;
            case '*':
                result = num1 * num2;
                break;
            case '/':
                result = num1 / num2;
                break;
        }
        textField.setText(String.valueOf(result));
        input = String.valueOf(result);
    } else if (command.equals("C")) {
        input = "";
        textField.setText("");
    }
}
public static void main(String[] args) {
    SwingUtilities.invokeLater(() -> new calc());
}
}
```

OUTPUT:



RESULT:

SOURCE CODE:

MyServer.java:

```
import java.net.*;
import java.io.*;
class MyServer {
    public static void main(String args[]) throws Exception {
        ServerSocket ss = new ServerSocket(3333);
        Socket s = ss.accept();
        DataInputStream din = new DataInputStream(s.getInputStream());
        DataOutputStream dout = new DataOutputStream(s.getOutputStream());
        BufferedReader br = new BufferedReader(new InputStreamReader(System.in));
        String str = "", str2 = "";
        while (!str.equals("stop")) {
            str = din.readUTF();
            System.out.println("Client says: " + str);
            str2 = br.readLine();
            dout.writeUTF(str2);
            dout.flush();
        }
        din.close();
        s.close();
        ss.close();
    }
}
```

MyClient.java:

```
import java.net.*;
import java.io.*;
class MyClient {
    public static void main(String args[]) throws Exception {
        Socket s = new Socket("localhost", 3333);
```

```
        DataInputStream din = new DataInputStream(s.getInputStream());
        DataOutputStream dout = new DataOutputStream(s.getOutputStream());
        BufferedReader br = new BufferedReader(new InputStreamReader(System.in));
        String str = "", str2 = "";
        while (!str.equals("stop")) {
            str = br.readLine();
            dout.writeUTF(str);
            dout.flush();
            str2 = din.readUTF();
            System.out.println("Server says: " + str2);
        }
        dout.close();
        s.close();
    }
}
```

OUTPUT:

```
Client sr/bin/env /usr/lib/jvm/java-11-openjdk-amd64/bin/java -cp /home/assassinvijay73/.vscode-server/data/User/wo
rkspaceStorage/6a9f2770ff347152d04f61bf0b61955b/redhat.java/jdt_ws/htdocs_e42e8491/bin My
HELLO
Server says:HELLO
I  AM  CLIENT
Server says: I AM  SERVER
```

```
Server sr/bin/env /usr/lib/jvm/java-11-openjdk-amd64/bin/java -cp /home/assassinvijay73/.vscode-server/data/User/wo
rkspaceStorage/6a9f2770ff347152d04f61bf0b61955b/redhat.java/jdt_ws/htdocs_e42e8491/bin My
HELLO
client says:HELLO
 I AM  SERVER
client says:I  AM  CLIENT
```

RESULT:

SOURCE CODE:

Client.java:

```
import java.rmi.Naming;

public class Client {

    public static void main(String[] args) {

        try {

            MyRemoteInterface remoteObject =

                (MyRemoteInterface) Naming.lookup("rmi://localhost/MyRemoteObject");

            String result = remoteObject.sendMessage("Hello from RMI Client!");

            System.out.println("Server Response: " + result);

        } catch (Exception e) {

            e.printStackTrace();

        }

    }

}
```

Server.java:

```
import java.rmi.Naming;

import java.rmi.registry.LocateRegistry;

public class Server {

public static void main(String[] args) {

        try {

            MyRemoteInterface remoteObject = new ServerImple();

            LocateRegistry.createRegistry(1099);

            Naming.rebind("MyRemoteObject", remoteObject);

            System.out.println("RMI Server is running...");
```

```
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

MyRemoteInterface.java:

```
import java.rmi.Remote;
import java.rmi.RemoteException;
public interface MyRemoteInterface extends Remote {
    String sayHello() throws RemoteException;
    String sendMessage(String message) throws RemoteException;
}
```

ServerImple.java:

```
import java.rmi.RemoteException;
import java.rmi.server.UnicastRemoteObject;
public class ServerImple extends UnicastRemoteObject implements MyRemoteInterface {
    public ServerImple() throws RemoteException {
        super();
    }
    @Override
    public String sayHello() throws RemoteException {
        return "Hello from RMI Server!";
    }
    @Override
    public String sendMessage(String message) throws RemoteException {
        return "Message received by the server: " + message;
    }
}
```

OUTPUT:

```
PS C:\Users\assas\Downloads\RMI (1)>  & 'C:\Program Files\Java\jdk-21\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:50085' '-XX:+ShowCodeDetailsInExceptionMessages' '-
cp' 'C:\Users\assas\AppData\Roaming\Code\User\workspaceStorage\8164192c3e85e7ee91753177ba6e5827\redhat.java\jdt_ws\RMI (1)_ef11b9cf\bin' 'Server'
RMI Server is running...
```

```
PS C:\Users\assas\Downloads\RMI (1)>  & 'C:\Program Files\Java\jdk-21\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:50099' '-XX:+ShowCodeDetailsInExceptionMessages' '-
cp' 'C:\Users\assas\AppData\Roaming\Code\User\workspaceStorage\8164192c3e85e7ee91753177ba6e5827\redhat.java\jdt_ws\RMI (1)_ef11b9cf\bin' 'Client'
Server Response: Message received by the server: Hello from RMI Client!
PS C:\Users\assas\Downloads\RMI (1)>
```

RESULT:

SOURCE CODE:

calcbean.java:

```
package SessionBean;

import javax.ejb.Stateless;

@Stateless

public class calcbean implements calcbeanLocal {

    @Override

    public Integer addition(int a, int b) {

        return (a + b);

    }

}
```

calcservlet.java:

```
package SessionBean;

import SessionBean.calcbeanLocal;

import java.io.IOException;

import java.io.PrintWriter;

import javax.ejb.EJB;

import javax.servlet.ServletException;

import javax.servlet.http.HttpServlet;

import javax.servlet.http.HttpServletRequest;

import javax.servlet.http.HttpServletResponse;

public class calcservlet extends HttpServlet {

    @EJB

    private calcbeanLocal calcbean;

    protected void processRequest(HttpServletRequest request, HttpServletResponse response)
```

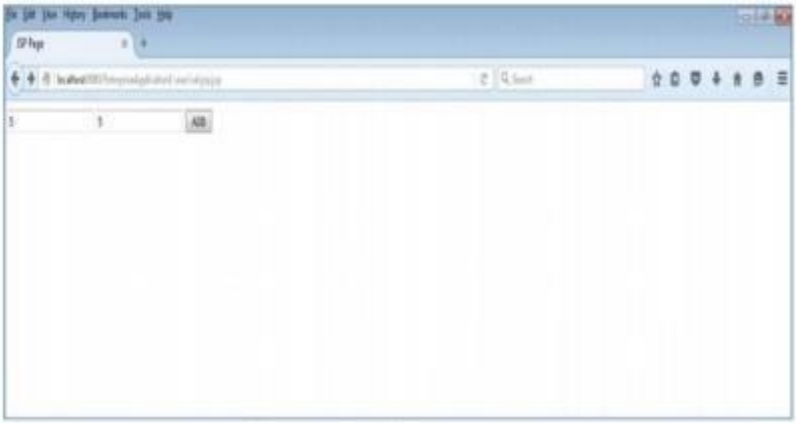
```
        throws ServletException, IOException {
    response.setContentType("text/html;charset=UTF-8");
    try (PrintWriter out = response.getWriter()) {
        out.println("<!DOCTYPE html>");
        out.println("<html>");
        out.println("<head>");
        out.println("<title>Servlet calcservlet</title>");
        out.println("</head>");
        out.println("<body>");
        int a = Integer.parseInt(request.getParameter("a1"));
        int b = Integer.parseInt(request.getParameter("b1"));
        out.println("<h1>SUM = " + calcbean.addition(a, b) + "</h1>");
        out.println("</body>");
        out.println("</html>");
    }
}
@Override
protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
    processRequest(request, response);
}
@Override
protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
    processRequest(request, response);
```

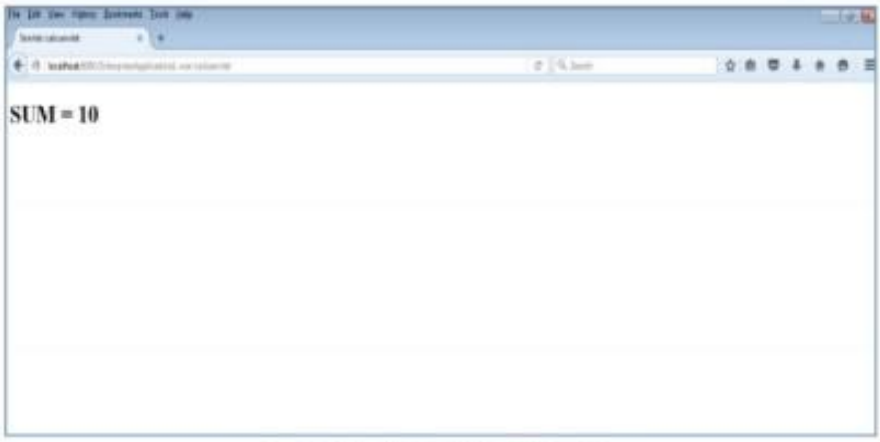
```
    }

    @Override
    public String getServletInfo() {
        return "Short description";
    }
}
```

calcjsp.jsp:

```
<!DOCTYPE html>
<html>
<head>
    <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
    <title>JSP Page</title>
</head>
<body>
    <form action="calcservlet" method="POST">
        Number 1: <input type="text" name="a1"><br><br>
        Number 2: <input type="text" name="b1"><br><br>
        <input type="submit" value="ADD">
    </form>
</body>
</html>
```

OUTPUT:





RESULT:

SOURCE CODE:

JavaApplication2.java

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.Statement;
public class JavaApplication2 {
    public static void main(String[] args) {
        Connection c = null;
        Statement stmt = null;
        try {
            // Load PostgreSQL JDBC driver
            Class.forName("org.postgresql.Driver");
            c = DriverManager.getConnection(
                    "jdbc:postgresql://localhost:5432/emp",
                    "postgres",
                    "Periyar"
            );
            c.setAutoCommit(false);
            System.out.println("Open Database Connection Successfully");
            stmt = c.createStatement();
            ResultSet rs = stmt.executeQuery("SELECT * FROM emp_detail;");
            while (rs.next()) {
                int id = rs.getInt("eid");
                String name = rs.getString("ename");
                int age = rs.getInt("eage");
                String address = rs.getString("eaddress");
                float salary = rs.getFloat("esalary");
```

```
            System.out.println("ID = " + id);
            System.out.println("NAME = " + name);
            System.out.println("AGE = " + age);
            System.out.println("ADDRESS = " + address);
            System.out.println("SALARY = " + salary);
            System.out.println();
        }
        rs.close();
        stmt.close();
        c.close();
    } catch (Exception e) {
        System.err.println(e.getClass().getName() + ": " + e.getMessage());
        System.exit(0);
    }
    System.out.println("Operation Done Successfully");
  }
}
```

OUTPUT:

```
Open Database Connection Successfully
ID = 1
NAME = Mr.X
AGE = 30
ADDRESS = Salem, Chennai
SALARY = 50000.0

ID = 2
NAME = Mr.X
AGE = 25
ADDRESS = Salem, Chennai
SALARY = 55000.0

Operation Done Successfully
```

RESULT:

SOURCE CODE:

PersonBean.java:

```
public class PersonBean {
    private String name;
    private int age;
    private String address;
    public String getName() {
        return name;
    }
    public void setName(String name) {
        this.name = name;
    }
    public int getAge() {
        return age;
    }
    public void setAge(int age) {
        this.age = age;
    }
    public String getAddress() {
        return address;
    }
    public void setAddress(String address) {
        this.address = address;
    }
}
```

JavaBean.java:

```
public class JavaBean {
    public static void main(String[] args) {
```

```
        PersonBean person = new PersonBean();
        person.setName("Mr.V");
        person.setAge(21);
        person.setAddress("SALEM");
        System.out.println("Name: " + person.getName());
        System.out.println("Age: " + person.getAge());
        System.out.println("Address: " + person.getAddress());
    }
}
```

OUTPUT:

```
PS C:\Users\assas\Desktop\New folder\java (1)> c:; cd 'c:\Users\assas\Desktop\New folder\java (1)'; & 'C:\Program Files\Java\jdk-21\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,addres
s=localhost:50360' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\assas\AppData\Roaming\Code\User\workspaceStorage\dfd99a7e5f0de3ee4429407099cad022\redhat.java\jdt_ws\java (1)_24fdad5\bin' 'JavaBea
n'
Name: Mr.V
Age: 21
Address: SALEM
```

RESULT: